Experimental Design & Scope of Inference

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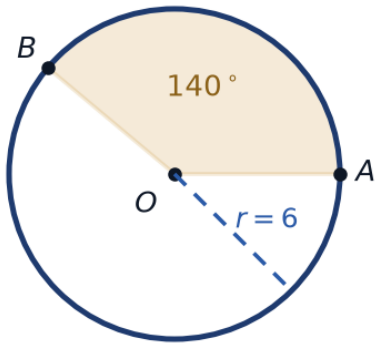
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Observational studies vs. experiments

A researcher records the exercise habits and cholesterol levels of 500 people without intervening in their lifestyle. Identify whether this is an observational study or an experiment, and explain what type of conclusion (association or causation) it can support.

2 [Foundational] • Target: Observational studies vs. experiments

A researcher randomly assigns 200 volunteers to either an exercise program or a no-exercise control group, then measures cholesterol after 6 months. Identify this design, and explain why it can support a causal conclusion.



3 [Foundational] • Target: The three principles of experimental design (RRCB)

State the three key principles of a well-designed experiment (sometimes remembered as RRC: Random assignment, Replication, Control of confounding variables), and briefly explain the purpose of each.

4 [Foundational] • Target: The three principles of experimental design (RRCB)

A study testing a new fertilizer randomly assigns 60 plants to fertilizer or no-fertilizer groups, but only uses 3 plants per group. Identify the design flaw, and explain the consequence.

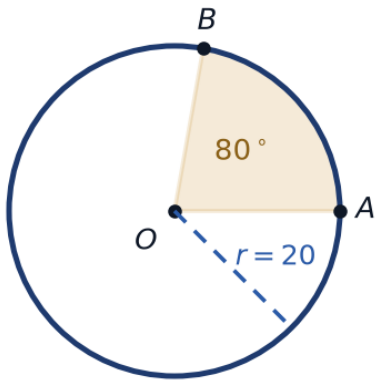
5 [Foundational] • Target: Blocking and matched pairs designs

A study on a new medication expects men and women to respond differently. Describe how a randomized block design would be set up, using sex as the blocking variable.

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6 [Foundational] • Target: Blocking and matched pairs designs

A researcher wants to compare two brands of running shoes on the same 20 runners, having each runner test both brands (in random order) and comparing their times. Identify this design.



7 [Foundational] • Target: Placebo effect and blinding

Explain what the placebo effect is, and describe how a single-blind experiment design controls for it.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Placebo effect and blinding

Explain the difference between a single-blind and a double-blind experiment.

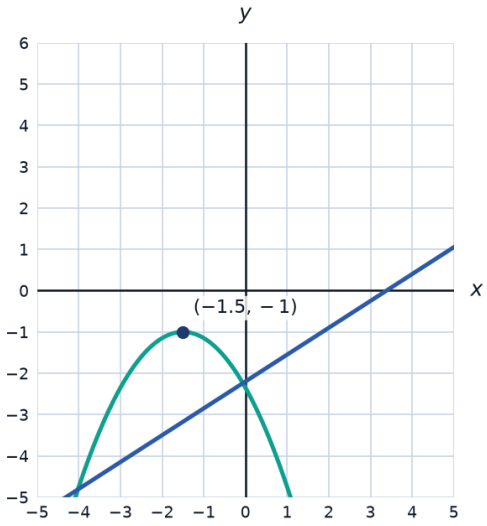
9 [Intermediate] • Target: Scope of inference

A study uses a simple random sample from the population but is purely observational (no random assignment of treatment). State what type of conclusion (about the population, about causation, both, or neither) is justified, and explain why.

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10 [Intermediate] • Target: Scope of inference

A study randomly assigns treatments to a group of volunteers who were not randomly selected from any larger population. State what type of conclusion is justified.



11 [Intermediate] • Target: Putting it together: evaluating a study design

A pharmaceutical company recruits 300 volunteers (not randomly selected from the general population) and randomly assigns them to receive either a new drug or a placebo, in a double-blind design. Evaluate what conclusions can and cannot be drawn from this study.

12 [Intermediate] • Target: Confounding variables

An observational study finds that people who drink more coffee have a higher rate of heart disease. Explain how smoking could act as a confounding variable in this relationship.

13 [Intermediate] • Target: Confounding variables

Explain how a well-designed randomized experiment would help rule out a variable like smoking as a confounder in the coffee and heart disease example.

14 [Intermediate] • Target: Identifying experimental units and treatments

A researcher tests three different fertilizer amounts (low, medium, high) on tomato plants, measuring final yield. Identify the experimental units, the treatment, and the response variable.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

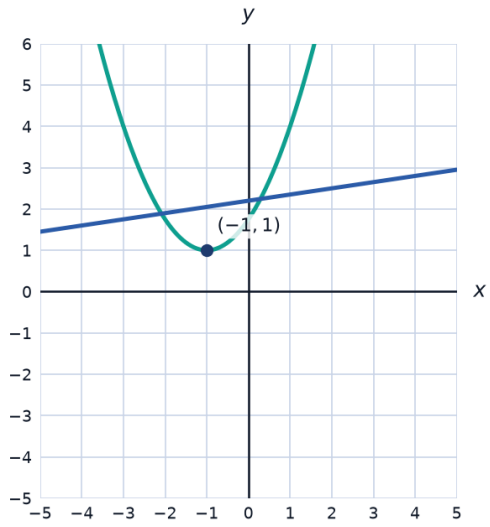
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15 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A pharmaceutical researcher is testing whether a new sleep aid reduces the time it takes people to fall asleep, and worries that people with severe insomnia might respond very differently from people with mild insomnia. Design a study using a randomized block design that accounts for this concern, describing each step, and explain the specific benefit this design provides over a simple completely randomized design in this scenario.

16 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A company tests a new website design by randomly assigning visitors to see either the old design or the new design, without telling visitors which version is the "experimental" one, and measures the percentage who complete a purchase. Explain why this study can support a causal conclusion about the website redesign, and describe the specific scope (which population) that conclusion applies to.



17 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 1.01 Measures of Center & Spread

A university studies whether a new tutoring program improves final exam scores. Volunteers who sign up for tutoring are compared to students who did not sign up, without any random assignment. At the end of the semester, tutored students score noticeably higher on average. Explain why this study cannot establish that the tutoring program caused the improvement, identifying a specific plausible confounding variable.

18 [Challenging] • Target: The three principles of experimental design (RRCB) • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

An agronomist wants to test whether a new organic fertilizer increases tomato yield compared to a standard commercial fertilizer. Explain how the agronomist must incorporate the four principles of experimental design: Comparison, Random Assignment, Control, and Replication.

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19 [Challenging] • Target: Blocking and matched pairs designs • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

A health study surveys 5,000 adults and finds that people who sleep 8 hours per night have significantly lower blood pressure than those who sleep 5 hours. (a) Can researchers conclude that increasing sleep causes a reduction in blood pressure? (b) Delineate the scope of inference using the 2×2 selection/assignment matrix.

20 [Challenging] • Target: Placebo effect and blinding • Cross-Lesson: 2.01 Sampling Methods & Bias

Level 3 · AdvancedAn investigator conducts a hypothesis test at significance level $\alpha = 0.05$. Explain two distinct ways the investigator could increase the statistical power of the test without changing the true effect size.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Observational studies vs. experiments

A researcher randomly assigns 467 volunteers to receive a daily supplement or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Observational study

b. Census

c. Experiment

d. Meta-analysis

22 [Foundational] • Target: The three principles of experimental design (RRCB)

A researcher records data on exercise habits and cholesterol levels from 435 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study

b. Experiment

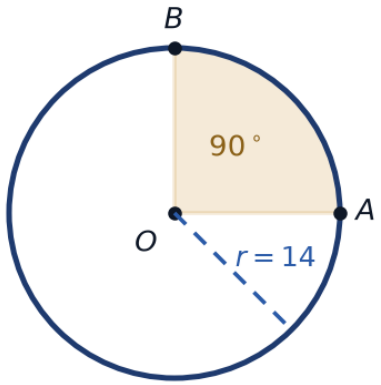
c. Meta-analysis

d. Census

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23 [Foundational] • Target: Blocking and matched pairs designs

A researcher randomly assigns 208 volunteers to receive a new exercise program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.



a. Observational study

b. Census

c. Experiment

d. Meta-analysis

24 [Foundational] • Target: Placebo effect and blinding

A researcher records data on fertilizer use and crop yield from 438 people without intervening in their lifestyle in any way. Identify the type of study.

a. Observational study

b. Experiment

c. Census

d. Meta-analysis

25 [Foundational] • Target: Scope of inference

A researcher randomly assigns 259 volunteers to receive a new medication or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Census

b. Meta-analysis

c. Observational study

d. Experiment

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26 [Foundational] • Target: Putting it together: evaluating a study design

A researcher randomly assigns 108 volunteers to receive a meditation app or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Experiment

b. Census

c. Meta-analysis

d. Observational study

27 [Foundational] • Target: Confounding variables

A researcher randomly assigns 476 volunteers to receive a tutoring program or serve as a control group, then measures the outcome after several weeks. Identify the type of study.

a. Census

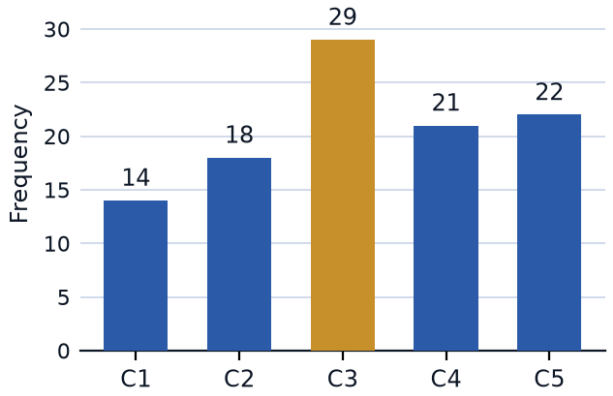
b. Meta-analysis

c. Observational study

d. Experiment

28 [Foundational] • Target: Identifying experimental units and treatments

A researcher records data on medication use and symptom severity from 391 people without intervening in their lifestyle in any way. Identify the type of study.



a. Observational study

b. Experiment

c. Census

d. Meta-analysis

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Observational studies vs. experiments

A researcher records data on meditation habits and stress levels from 301 people without intervening in their lifestyle in any way. Identify the type of study.

a. Experiment

b. Observational study

c. Census

d. Meta-analysis

30 [Intermediate] • Target: The three principles of experimental design (RRCB)

In a study on medication use and symptom severity, a simple random sample of 130 people is selected from the population of interest; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population

c. Association only, and only about the subjects actually studied (not generalizable)

d. Association only, generalizable to the entire population

31 [Intermediate] • Target: Blocking and matched pairs designs

In a study on fertilizer use and crop yield, 58 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a new fertilizer or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Association only, generalizable to the entire population

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

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32 [Intermediate] • Target: Placebo effect and blinding

In a study on tutoring attendance and exam performance, a simple random sample of 361 people is selected from the population of interest; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

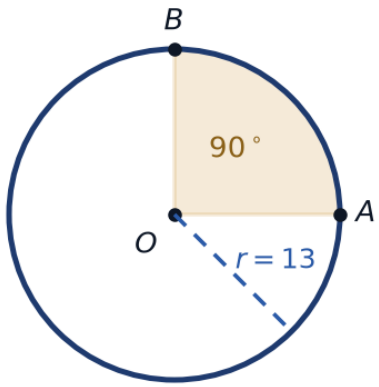
b. Association only, generalizable to the entire population

c. Causation, but only about the subjects actually studied (not generalizable)

d. Association only, and only about the subjects actually studied (not generalizable)

33 [Intermediate] • Target: Scope of inference

In a study on tutoring attendance and exam performance, 127 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a tutoring program or not, without assigning anything. What type of conclusion is justified by this design?



a. Association only, generalizable to the entire population

b. Causation, but only about the subjects actually studied (not generalizable)

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

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34 [Intermediate] • Target: Putting it together: evaluating a study design

In a study on tutoring attendance and exam performance, a simple random sample of 275 people is selected from the population of interest; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, generalizable to the entire population

c. Association only, and only about the subjects actually studied (not generalizable)

d. Causation, but only about the subjects actually studied (not generalizable)

35 [Intermediate] • Target: Confounding variables

In a study on tutoring attendance and exam performance, 459 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a tutoring program or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Association only, generalizable to the entire population

c. Causation, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

36 [Intermediate] • Target: Identifying experimental units and treatments

In a study on medication use and symptom severity, 471 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a new medication or a control condition. What type of conclusion is justified by this design?

a. Association only, and only about the subjects actually studied (not generalizable)

b. Causation, but only about the subjects actually studied (not generalizable)

c. Causation, generalizable to the entire population

d. Association only, generalizable to the entire population

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37 **[Intermediate]** • **Target: Observational studies vs. experiments**

In a study on medication use and symptom severity, a simple random sample of 476 people is selected from the population of interest; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, generalizable to the entire population

c. Association only, and only about the subjects actually studied (not generalizable)

d. Causation, but only about the subjects actually studied (not generalizable)

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: The three principles of experimental design (RRCB)** • **Cross-Lesson: 1.02 Z-Scores & Linear Transformations**

In a study on medication use and symptom severity, 165 volunteers (not randomly selected from any larger population) take part; researchers simply record whether each person already uses a new medication or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, and only about the subjects actually studied (not generalizable)

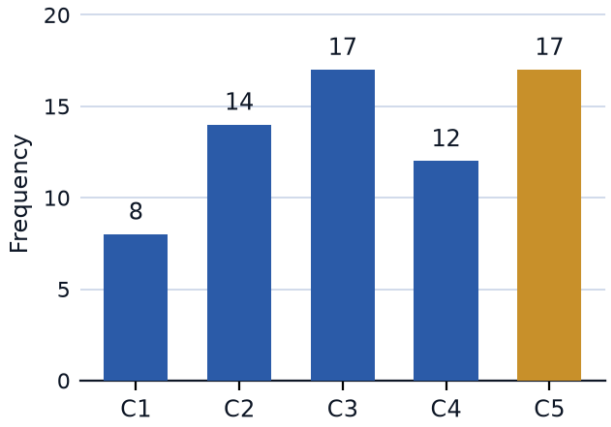
c. Association only, generalizable to the entire population

d. Causation, but only about the subjects actually studied (not generalizable)

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39 [Challenging] • Target: Blocking and matched pairs designs • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

In a study on supplement use and energy levels, a simple random sample of 406 people is selected from the population of interest; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?



a. Association only, and only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population

c. Association only, generalizable to the entire population

d. Causation, but only about the subjects actually studied (not generalizable)

40 [Challenging] • Target: Placebo effect and blinding • Cross-Lesson: 2.01 Sampling Methods & Bias

In a study on supplement use and energy levels, 172 volunteers (not randomly selected from any larger population) take part; they are then randomly assigned to receive a daily supplement or a control condition. What type of conclusion is justified by this design?

a. Causation, but only about the subjects actually studied (not generalizable)

b. Causation, generalizable to the entire population

c. Association only, generalizable to the entire population

d. Association only, and only about the subjects actually studied (not generalizable)

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41 [Challenging] • Target: Scope of inference • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

In a study on fertilizer use and crop yield, a simple random sample of 192 people is selected from the population of interest; researchers simply record whether each person already uses a new fertilizer or not, without assigning anything. What type of conclusion is justified by this design?

a. Causation, generalizable to the entire population

b. Association only, generalizable to the entire population

c. Causation, but only about the subjects actually studied (not generalizable)

d. Association only, and only about the subjects actually studied (not generalizable)

42 [Challenging] • Target: Putting it together: evaluating a study design • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

In a study testing a new medication, subjects know whether they are receiving the real treatment or a placebo, and so do the researchers evaluating the results. Identify the blinding used.

a. Not blinded

b. Double-blind

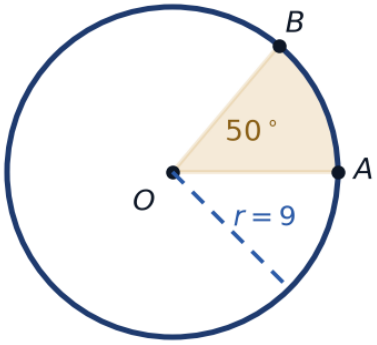
c. Single-blind

d. Triple-blind

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43 [Challenging] • Target: Confounding variables • Cross-Lesson: 3.03 The Normal Distribution

In a study testing a new fertilizer, subjects do not know whether they are receiving the real treatment or a placebo, but the researchers evaluating the results do know which group each subject is in. Identify the blinding used.



a. Not blinded

b. Single-blind

c. Triple-blind

d. Double-blind

44 [Challenging] • Target: Identifying experimental units and treatments • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

In a study testing a new fertilizer, neither the subjects nor the researchers evaluating the results know which subjects received the real treatment versus a placebo. Identify the blinding used.

a. Double-blind

b. Triple-blind

c. Single-blind

d. Not blinded

45 [Challenging] • Target: Observational studies vs. experiments • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

In a study testing a new exercise program, subjects know whether they are receiving the real treatment or a placebo, and so do the researchers evaluating the results. Identify the blinding used.

a. Double-blind

b. Single-blind

c. Not blinded

d. Triple-blind

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